Multi-Agent Negotiation for Right-of-Way in Complex Intersections

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***Abstract*—Autonomous vehicles operating at complex uncontrolled intersections must identify conflicting movements, infer surrounding-vehicle intentions, obey right-of-way rules, and coordinate with other vehicles using limited local information. This paper presents a decentralized, rule-aware multi-agent framework integrating ego-local perception, local dynamic mapping, two-stage Gated Recurrent Unit (GRU) intention prediction, physical conflict-zone reasoning, traffic-rule precedence, graph-based interaction representation, and Multi-Agent Proximal Policy Optimization (MAPPO) for right-of-way negotiation. Mandatory regulatory and physical safety constraints remain outside the learned policy and restrict invalid negotiation actions. On the held-out inD test recording, the 1.0 s and 0.5 s intention models achieved accuracies of 89.66% and 93.10%, respectively. Across three MAPPO training replications, mean aggregate team travel time decreased descriptively from 17,064.12 to 15,846.64 vehicle-s. No collisions or blocked-zone violations were recorded in the reported training and demonstration runs. An end-to-end evaluation using the Simulation of Urban MObility (SUMO) also identified a remaining liveness limitation when a precedence cycle was not resolved. The results demonstrate the feasibility of the proposed negotiation architecture while motivating multi-round negotiation and broader held-out evaluation.**

***Index Terms*—autonomous vehicles, multi-agent reinforcement learning, right-of-way negotiation, intention prediction, unsignalized intersections, SUMO**

I. Introduction

Autonomous vehicles (AVs) require reliable decision-making mechanisms to interact safely with other vehicles in complex traffic environments. Intersections are particularly challenging because multiple vehicles may arrive from different directions with different intentions, such as travelling straight or turning, which can create competing or conflicting movements. Therefore, an AV must identify relevant conflicts and determine appropriate right-of-way relationships before entering the intersection [1].

Recent research has increasingly investigated learning-based and right-of-way-aware coordination for AVs at unsignalized intersections. Multi-agent reinforcement learning (MARL) has demonstrated potential for cooperative vehicle coordination [2], [3], while recent right-of-way-aware MARL approaches explicitly incorporate precedence information into learned decision-making [1]. Nevertheless, robust decision-making remains challenging under partial observations, uncertain vehicle intentions, multiple simultaneous conflicts, and safety-critical right-of-way constraints [4].

This research proposes a **multi-agent negotiation framework for right-of-way in complex intersections**. Each AV independently constructs an ego-local representation of surrounding traffic using perception and a Local Dynamic Map (LDM). A Gated Recurrent Unit (GRU)-based intention model estimates the likely movement of relevant vehicles [5]. Predicted movements are combined with map-based conflict zones, temporal and kinematic reasoning, and traffic-rule precedence to identify meaningful right-of-way conflicts. The resulting interaction is represented as a graph and supplied to role-specific Multi-Agent Proximal Policy Optimization (MAPPO) policies operating under centralized training and decentralized execution [6].

The learned negotiation mechanism does not directly control vehicle motion or override mandatory traffic rules. Instead, it

negotiates eligible precedence relationships under action constraints. The negotiated result is converted into conflict-zone-aware crossing obligations and executable vehicle-control decisions in the Simulation of Urban MObility (SUMO) environment [7]. This separation allows learned negotiation to operate within deterministic traffic-rule and safety constraints.

The main contributions of this research are:

1) An **ego-local right-of-way reasoning framework** integrating perception, intention prediction, conflict analysis, temporal reasoning, and traffic-rule precedence.
2) A **rule-constrained graph-based multi-agent negotiation mechanism** using role-specific MAPPO policies for decentralized right-of-way decisions.
3) A **negotiation-to-execution mechanism** that converts negotiated precedence relationships into conflict-zone-aware vehicle behaviour.
4) An **end-to-end SUMO-based evaluation framework** for studying multi-vehicle right-of-way negotiation at complex intersections.

The proposed framework is evaluated using a simulated four-way unsignalized intersection containing multiple AVs performing straight, left-turn, and right-turn movements. The experiments evaluate the individual reasoning components and the complete negotiation pipeline to examine the feasibility of the proposed approach.

II. Related Work

Autonomous intersection management has increasingly focused on coordinating connected and autonomous vehicles without conventional traffic signals. Recent work has investigated scheduling, trajectory coordination, and conflict-aware intersection management [8]. Feng *et al.* [1] incorporated explicit right-of-way guidance into a multi-agent deep reinforcement-learning framework for unsignalized intersections. These studies highlight the importance of combining interaction-aware coordination with safety and efficiency objectives.

MARL has also been applied directly to intersection coordination. Spatharis and Blekas [2] investigated multi-agent deep reinforcement learning for AVs operating in traffic networks containing unsignalized intersections. Zhuang *et al.* [3] proposed an attention-based MAPPO approach for cooperative decision-making at an unsignalized intersection. MAPPO has also demonstrated strong performance in cooperative multi-agent environments [6]. In contrast, the present work restricts learned negotiation using explicit traffic-rule precedence and conflict-based action constraints rather than allowing the learned policy to determine all right-of-way behaviour directly.

Vehicle-intention prediction is another important component of intersection decision-making. Komol *et al.* [5] investigated recurrent neural networks, including GRU-based models, for early prediction of intended vehicle movements at intersections. The inD dataset introduced by Bock *et al.* [9] provides naturalistic road-user trajectories from real German intersections and is used in this research for intention-model development. Graph-based approaches have also been investigated for interaction-aware prediction and coordination. Azadani and Boukerche [10] modelled vehicle interactions using spatio-temporal attention graphs, while Zhou *et al.* [11] combined graph-based interaction reasoning with reinforcement learning for unsignalized intersection coordination.

Existing research therefore provides important foundations in MARL, intention prediction, autonomous intersection management, and graph-based interaction modelling. The present research integrates ego-local observation, learned intention estimation, physical conflict-zone reasoning, explicit traffic-rule precedence, graph-based negotiation, and negotiation-to-execution logic within a single right-of-way framework. Learning is applied only where negotiation is permitted, while mandatory traffic and safety constraints remain outside the learned policy.

III. Methodology

A. System Architecture

The proposed framework performs decentralized right-of-way negotiation among AVs in a complex intersection. SUMO is used as the microscopic traffic simulator [7]. At each simulation step, every AV independently observes surrounding traffic, maintains its LDM, predicts the intentions of relevant vehicles, identifies physical conflicts, applies traffic-rule precedence, and constructs a local right-of-way graph. The graph is processed by the multi-agent negotiation policy, and the negotiated result is converted into conflict-zone-aware vehicle-control constraints.

A key design principle is the separation between *mandatory reasoning* and *learned negotiation*. Traffic rules, physical conflict relationships, and execution-safety constraints are determined outside the reinforcement-learning policy. Learning is used only for precedence relationships that are eligible for negotiation.

B. Perception and Intention Estimation

Each AV constructs its own LDM from ego-local observations rather than receiving a complete global traffic state. The perception interface applies sensing-range and field-of-view limits, ego-coordinate transformation, vehicle-dimension reasoning, partial visibility, and dynamic occlusion. Temporarily unobserved vehicles are propagated using their previous motion state, while stale tracks are eventually removed. Ground-truth route or maneuver information is not exposed to the ego vehicle.

Vehicle intention is estimated using two GRU models trained on naturalistic trajectories from the inD dataset [9]. For each eligible target, 50 observations sampled at 25 Hz provide a 2 s motion history. The primary and secondary models predict LEFT, RIGHT, or STRAIGHT at 1.0 s and 0.5 s lead times, respectively. Predictions are accepted only when their confidence exceeds validation-selected thresholds; otherwise, or when the two stages disagree, the intention is treated as UNKNOWN. Unknown intentions are handled conservatively by retaining all feasible candidate movements.

C. Conflict and Traffic-Rule Reasoning

Legal movement paths are extracted from the SUMO road network. Each path is converted into a swept vehicle region, and the conflict region between movement paths i and j is defined as

$$Z_{ij} = P_i \cap P_j, \tag{1}$$

where P_i and P_j denote the swept regions of the two paths. Only crossing or merging movements with physical overlap are treated as spatial conflicts. Temporal and kinematic reasoning further considers vehicle progress, expected conflict-zone entry and clearance times, reachability, and the ability to stop before a conflict zone. Recent conflict-free intersection-management research similarly uses conflict relationships to determine safe vehicle ordering [12].

For relevant conflicts, regulatory precedence is determined according to the German Road Traffic Regulations (*Straßenverkehrs-Ordnung*, StVO) [13], including right-before-left and turning-related yielding requirements. The resulting relationships are represented as a directed precedence graph, where an edge $A \rightarrow B$ indicates that vehicle A must yield to vehicle B. Local claims are exchanged using an idealized vehicle-to-vehicle (V2V) communication model and combined into each AV's joint local graph. Precedence cycles are detected using strongly connected components.

D. Graph-Based Negotiation and Execution

The joint precedence graph is encoded using an edge-aware graph neural network (GNN) with message passing. Recent graph-based reinforcement-learning research has demonstrated the suitability of GNN representations for complex vehicle interactions at unsignalized intersections [11]. The implementation uses a 64-dimensional hidden representation with three message-passing layers. The graph encoder is deterministically initialized and kept frozen during the reported MAPPO experiments.

Two semantic actor roles are used. A proposer selects either `KEEP_CLAIM` or `RELINQUISH_CLAIM`, while a responder selects either `ACCEPT_RELINQUISHMENT` or `REJECT_RELINQUISHMENT`. Invalid actions are removed using hard action masks [14]. A centralized critic is used only during training and is unavailable during decentralized runtime execution. The cooperative reward is based on reducing total vehicle travel time.

Learned actions do not directly generate acceleration or steering commands. A deterministic protocol validates the negotiation result and maps valid precedence relationships back to physical conflict-zone obligations. Vehicles with unresolved outgoing obligations are prevented from entering the corresponding conflict zones, while eligible vehicles are permitted to proceed through SUMO.

IV. Experimental Setup and Results

A. Experimental Setup

The proposed framework was evaluated using SUMO [7] on a four-way uncontrolled intersection. The simulation used a time step of 0.04 s and a maximum episode duration of 200 s. All vehicles were AVs with a maximum speed of 13.89 m/s, length of 5.0 m, width of 1.8 m, acceleration of 2.0 m/s^2, and normal deceleration of 4.5 m/s^2. Twelve legal vehicle movements were available, including straight, left-turn, and right-turn movements from each of the four approaches. The map-processing stage generated 30 physical conflict zones.

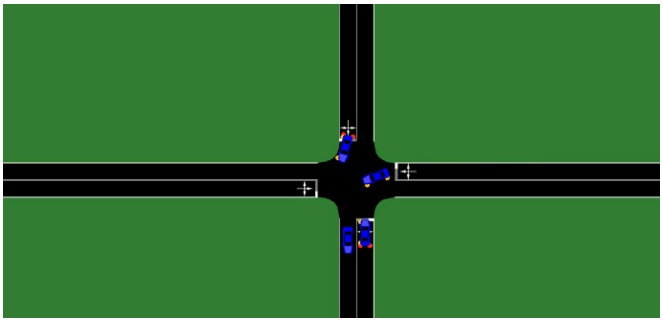


Fig. 1. SUMO four-way uncontrolled intersection used for multi-agent right-of-way negotiation experiments.

The intention-prediction models were trained using the inD intersection dataset [9]. Recordings 7–14 were used for training, recordings 15–16 for validation, and recording 17 was reserved for final testing. Each model used 50 position observations sampled at 25 Hz, corresponding to a 2 s observation history. The held-out test set contained 58 sequences for each prediction horizon. The validation-selected prediction-confidence thresholds were 0.4368 for the 1.0 s primary model and 0.3873 for the 0.5 s secondary model.

For negotiation learning, 108 regulatory-cycle scenarios were generated and divided into 36 training, 36 validation, and 36 held-out test scenarios without overlap. The reported MAPPO learning evidence was obtained from the training split using three independent replications. Two Proximal Policy Optimization (PPO) update cycles were performed per replication, with five PPO epochs per update. Adam optimization used a learning rate of 5×10^{-4} and a clipping parameter of 0.2. The graph encoder remained frozen during MAPPO training.

The simulated four-way intersection used for the experiments is shown in Fig. 1.

B. Intention-Prediction Test Results

The selected GRU models were evaluated once on the held-out recording 17 after model selection was completed using validation data. Table I compares the GRU models with the fixed weighted-logistic baseline.

TABLE I
HELD-OUT TEST PERFORMANCE OF THE INTENTION-PREDICTION MODELS.

Lead	Model	Accuracy	Bal. Acc.	Macro F1	Weighted F1
1.0 s	Weighted Logistic	79.31%	76.60%	77.46%	78.88%
1.0 s	GRU	**89.66%**	**89.10%**	**88.93%**	**89.83%**
0.5 s	Weighted Logistic	87.93%	86.22%	85.91%	87.68%
0.5 s	GRU	**93.10%**	**91.67%**	**92.38%**	**93.05%**

The primary GRU improved test accuracy from 79.31% for the weighted-logistic baseline to 89.66%, while the secondary GRU improved accuracy from 87.93% to 93.10%. The secondary 0.5 s model achieved the highest overall classification performance.

Table II presents the class-wise performance of the selected GRU models.

TABLE II
CLASS-WISE GRU PERFORMANCE ON THE HELD-OUT TEST SET.

Lead	Class	Precision	Recall	F1	Support
1.0 s	LEFT	93.33%	87.50%	90.32%	16
1.0 s	RIGHT	77.78%	87.50%	82.35%	16
1.0 s	STRAIGHT	96.00%	92.31%	94.12%	26
0.5 s	LEFT	87.50%	87.50%	87.50%	16
0.5 s	RIGHT	100.00%	87.50%	93.33%	16
0.5 s	STRAIGHT	92.86%	100.00%	96.30%	26

The exported Open Neural Network Exchange (ONNX) models were additionally checked against the original model outputs on all 58 test sequences. The maximum probability differences were approximately 1.79×10^{-7} and 1.34×10^{-7} for the primary and secondary models, respectively, confirming numerical consistency after deployment conversion.

C. MAPPO Training Results

Three independent MAPPO training replications were evaluated using aggregate team travel time, where lower values indicate lower collective delay.

TABLE III
MAPPO TRAINING RESULTS BASED ON AGGREGATE TEAM TRAVEL TIME.

Policy State	Mean Team Travel Time (vehicle-s)	Standard Deviation
C0	17,064.12	2,668.82
C1	16,205.36	4,297.67
C2	**15,846.64**	3,663.88

The mean team travel time decreased by 1,217.48 vehicle-s between C0 and C2. Across the three replications, 324 training-scenario executions and 1,620,000 SUMO simulation steps were recorded, with no recorded collisions or blocked-zone violations. These results indicate descriptive improvement during the limited training budget, but they do not establish convergence or statistical significance.

D. End-to-End Prototype Demonstration

A frozen MAPPO checkpoint was executed without further training in three structurally selected SUMO scenarios. Table IV summarizes the observed outcomes.

TABLE IV
END-TO-END MAPPO RESEARCH-PROTOTYPE DEMONSTRATION RESULTS.

Scenario	Prop.	Resp.	Completed	Travel Time (s)	Collisions
Regulatory cycle	8	3	2	490.76	0
Multi-factor negotiation	4	0	0	600.00	0
Nonphysical interpretation	4	3	1	453.44	0
Total	16	6	3	1544.20	0

The demonstration produced 22 learned MAPPO decisions, consisting of 16 proposer and 6 responder actions. No centralized critic or ground-truth route information was used during runtime, and no blocked-zone violations were recorded. However, the multi-factor scenario produced no completed vehicles because the remaining precedence cycle was not successfully resolved. This reveals a liveness limitation in the current prototype while also showing that unresolved conflicts are conservatively prevented from entering the intersection.

V. DISCUSSION

The experimental results indicate that the proposed framework can integrate intention prediction, conflict reasoning, traffic-rule precedence, and learned negotiation within a decentralized intersection-management pipeline. The GRU models achieved higher held-out test performance than the weighted-logistic baseline at both prediction horizons. The secondary 0.5 s model achieved the highest accuracy of 93.10% and Macro F1 of 92.38%, while the primary 1.0 s model achieved 89.66% accuracy. The shorter prediction horizon benefits from motion information closer to the actual maneuver, whereas the earlier 1.0 s prediction provides additional time for conflict reasoning and negotiation.

The MAPPO experiments showed a descriptive reduction in mean team travel time from 17,064.12 vehicle-s at C0 to 15,846.64 vehicle-s at C2 across three training replications. No collisions or blocked-zone violations were recorded during the 324 training-scenario executions. These observations indicate that negotiation behaviour changed during learning while the deterministic traffic-rule and execution layers continued to restrict unsafe actions. However, the limited number of PPO updates and replications is insufficient to establish convergence, statistical significance, or superiority over existing coordination approaches.

The end-to-end demonstration further showed that learned proposer and responder actions can be executed using ego-local observations without runtime access to the centralized critic or ground-truth routes. Nevertheless, an important liveness limitation was observed. In one multi-vehicle scenario, the negotiated precedence graph remained cyclic and no vehicle completed its movement. The safety mechanism prevented vehicles with unresolved precedence obligations from entering the conflict region, but this conservative behaviour can result in deadlock when negotiation does not produce an executable precedence order.

Several limitations remain. The MAPPO policy has not yet been evaluated on the reserved 36-scenario held-out test set, and no matched comparison with a rule-based or alternative negotiation baseline has been performed. Ablation experiments for the graph encoder, hard action masks, traffic-rule reasoning, and intention prediction are also not yet available. In addition, perception is represented using a geometric sensing abstraction rather than a complete physical sensor model. Future work should therefore investigate multi-round negotiation and deadlock recovery, followed by frozen held-out evaluation, matched baseline comparisons, ablation studies, and testing

under additional intersection configurations and perception conditions.

VI. Conclusion

This research presented a decentralized multi-agent framework for right-of-way negotiation in complex intersections. The proposed approach integrates ego-local perception, LDMs, GRU-based intention prediction, spatial and temporal conflict reasoning, traffic-rule precedence, graph-based representation, and MAPPO-based negotiation while keeping mandatory safety and regulatory constraints outside the learned policy.

The intention-prediction models achieved held-out test accuracies of 89.66% and 93.10% for the 1.0 s and 0.5 s prediction horizons, respectively. The MAPPO experiments also showed a descriptive reduction in mean team travel time while recording no collisions or blocked-zone violations in the reported training and demonstration runs. However, the end-to-end evaluation identified a remaining liveness limitation when unresolved precedence cycles prevent vehicles from progressing. Future work will focus on multi-round negotiation and deadlock recovery, followed by evaluation on the reserved held-out scenarios, baseline comparisons, ablation studies, and testing across additional intersection configurations and perception conditions.

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